

Virtual Machine Configuration and Robot Connection

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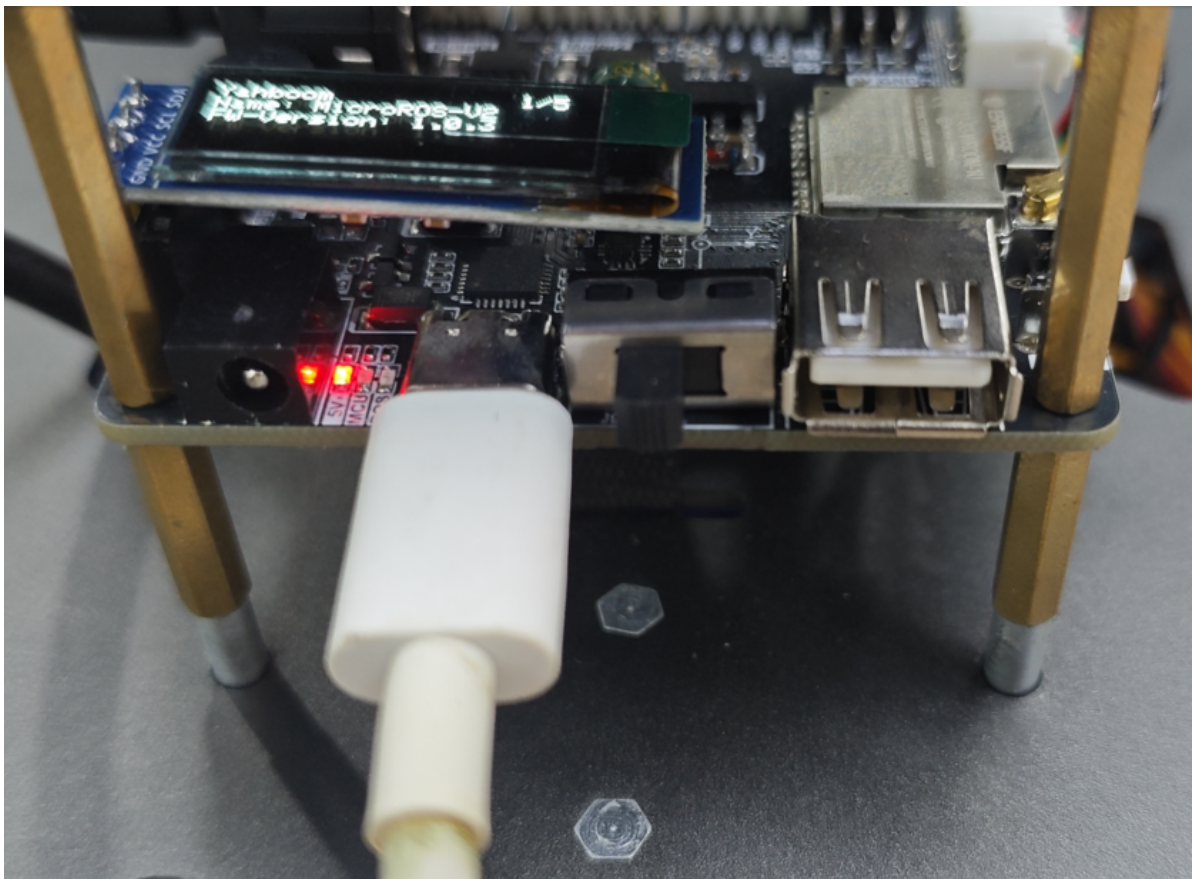
1. Course Content
2. Preparation
3. Configure Vehicle Controller Parameters
 - 3.1 Configure Control Board Parameters
 - 3.2 Configure the Vehicle Type and Camera IP in the Virtual Machine
4. Test the Connection

1. Course Content

- Connect the computer and the MicroROS-V2 robot via a USB data cable, and use the RobotManager control panel to configure the vehicle controller's parameters, such as the Microros agent IP address and servo offset

2. Preparation

- Start the factory-bundled virtual machine **MicroRosV2-VirtualMachine**
- Toggle the MicroRos control board power switch; after the device boots up, connect the computer and the vehicle controller using a **type-c** data cable

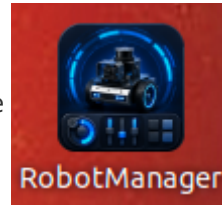


- In the device option window that pops up on the computer, choose to connect the USB device to the virtual machine

3. Configure Vehicle Controller Parameters

3.1 Configure Control Board Parameters

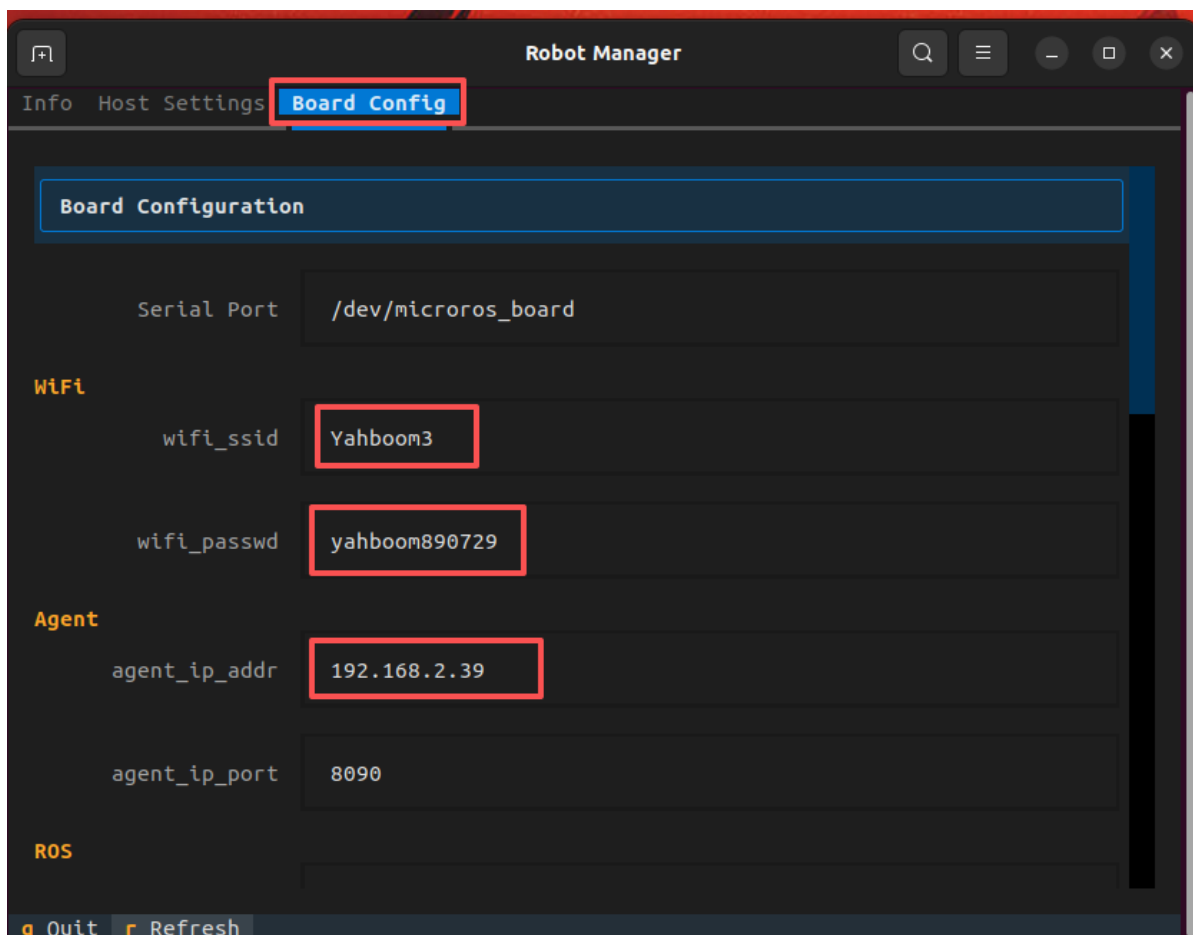
- On the virtual machine desktop, double-click the



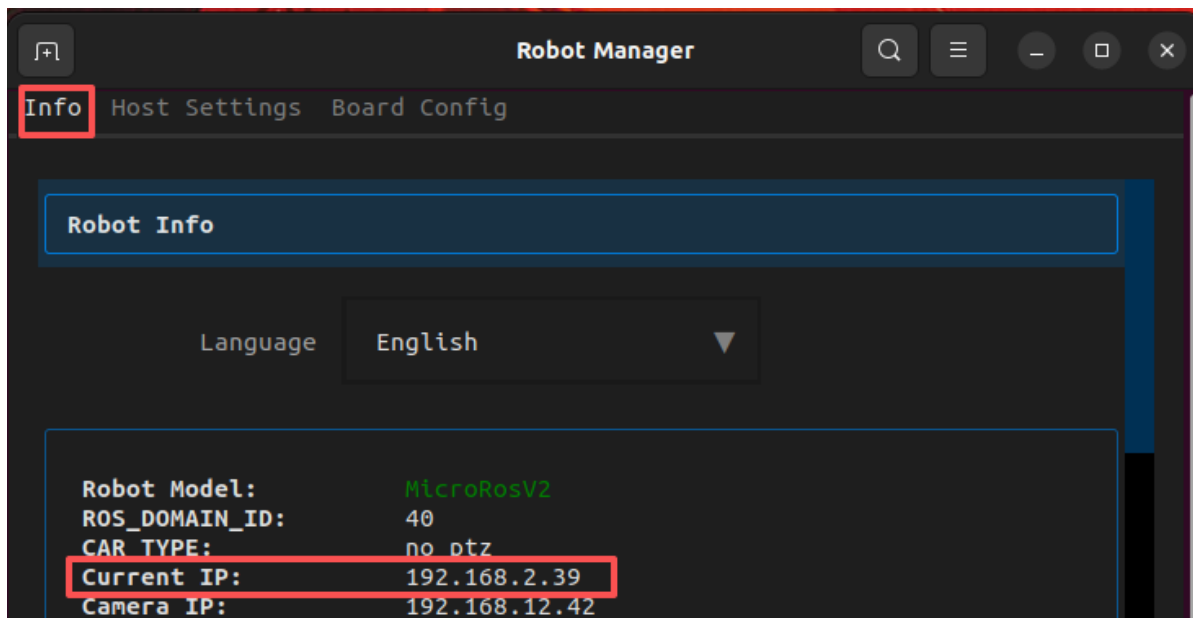
RobotManager

application icon

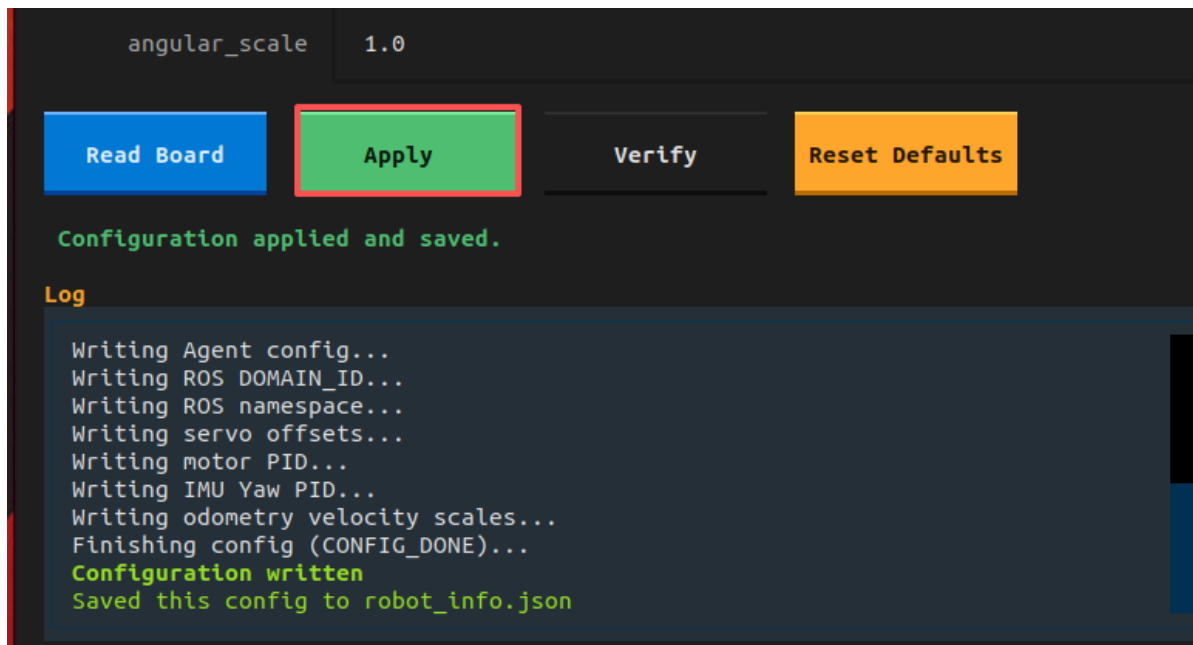
- Click the **Control Board** panel and enter the WiFi name and password (make sure the network is on the same subnet as the virtual machine)



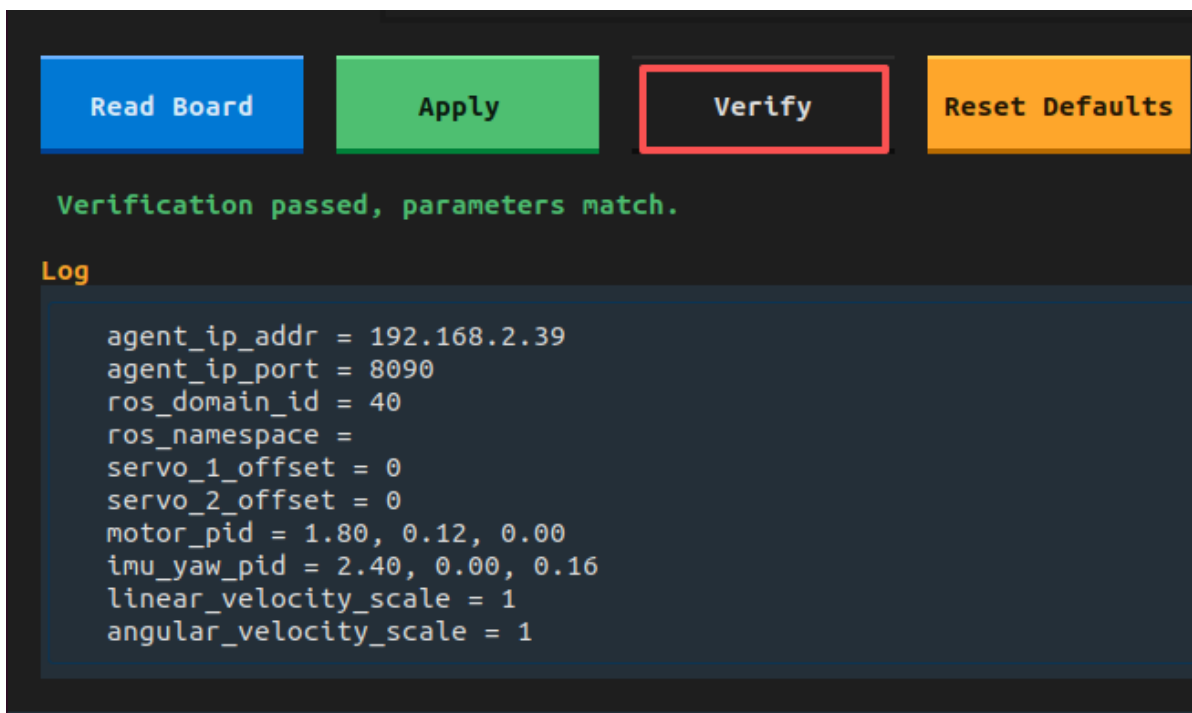
- Enter the virtual machine's IP address in `agent_ip_addr`; you can view the virtual machine's IP address in the Info panel



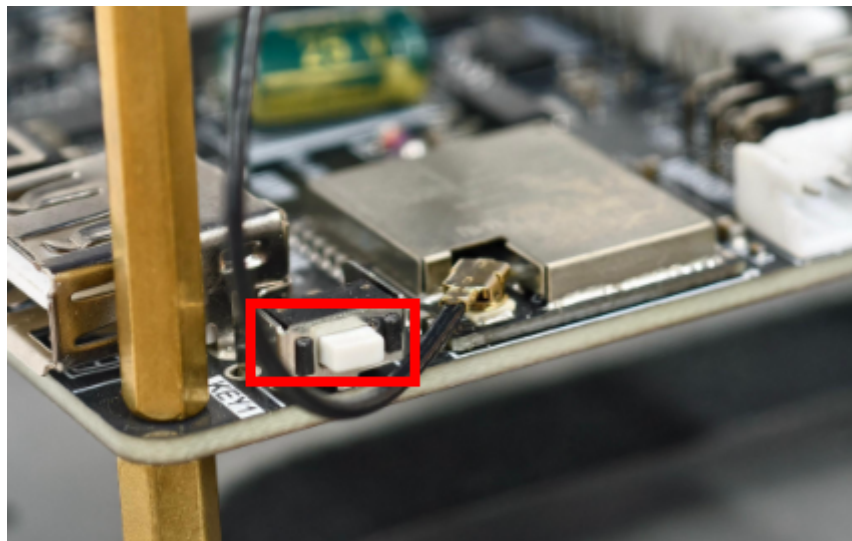
- Click Apply Config to write the above parameters into the MicroRos control board



- Click Verify; as long as the parameter check passes, the control board parameter configuration is complete



- Click the button on the right side of the control board to switch the OLED screen display content and view the information



[!IMPORTANT]

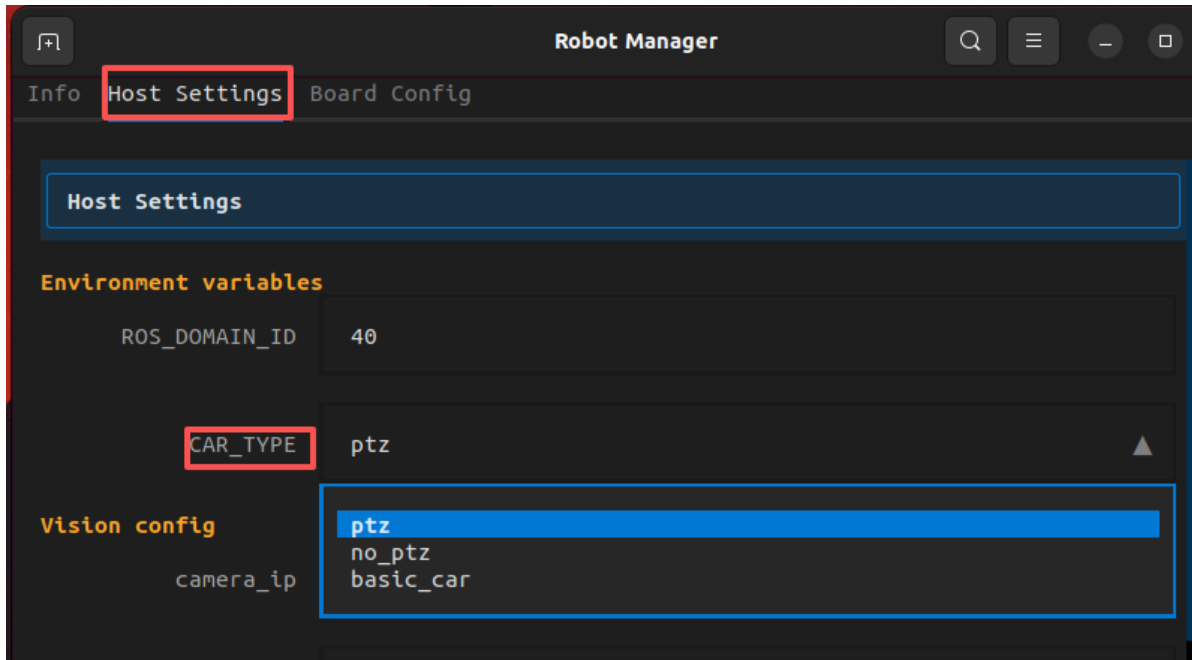
self: the vehicle controller's own IP address; make sure it is on the same subnet as the virtual machine

agent: the virtual machine IP address written in the previous step

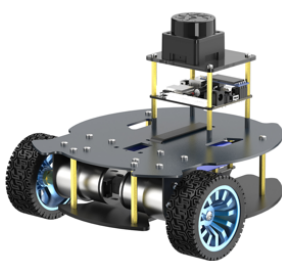
camera: the image-transmission camera's IP address

3.2 Configure the Vehicle Type and Camera IP in the Virtual Machine

- In the host settings, open the CAR_TYPE dropdown list and select the corresponding vehicle model



- The vehicle model names are as follows:



basic_car

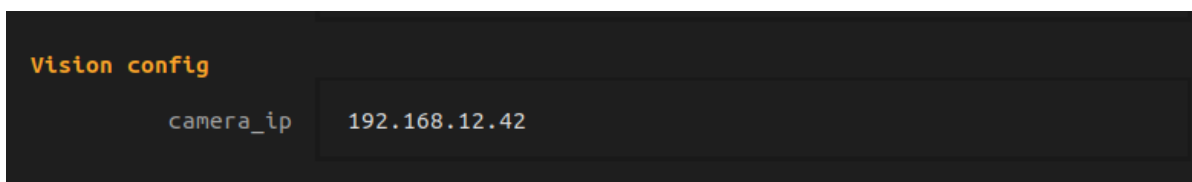


no_ptz

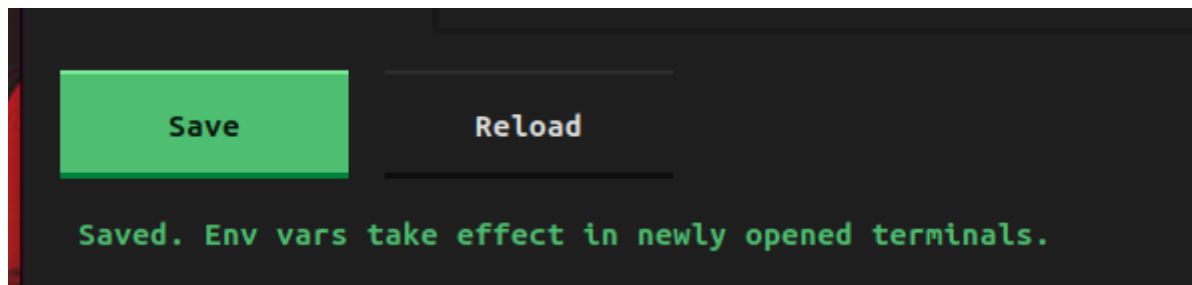


ptz

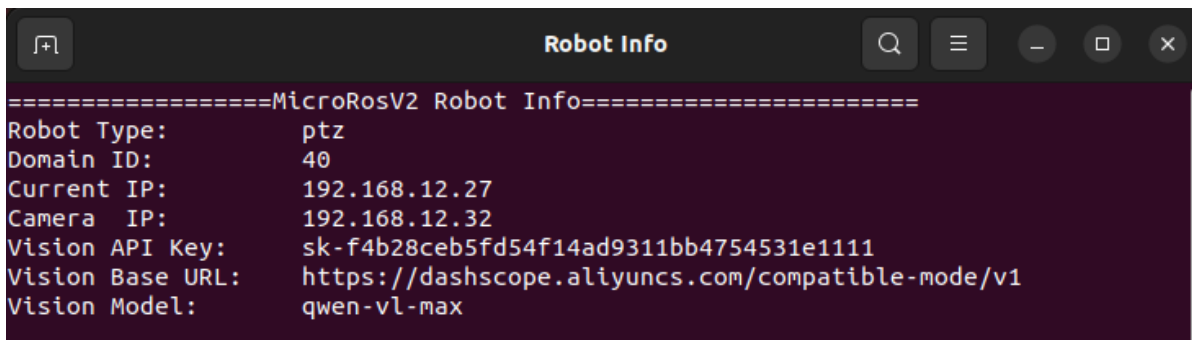
- Camera IP configuration: enter the camera IP address shown on the vehicle controller's OLED screen



- Click Save Config; press **q** or simply close the window to exit **RobotManager**



- Later, when you need to view endpoint configurations such as DomainID, host IP, and camera IP, click the Robot Info icon in the sidebar for a quick overview



4. Test the Connection

- After completing the above configuration, launch the example program to test whether the virtual machine and the robot's vehicle controller can communicate normally

```
ros2 launch microros_v2_bringup test_connect.launch.py
```

- The terminal window prints statistics for the IMU and LiDAR topics every 10 seconds, and the image window shows the camera feed, proving that communication between the virtual machine and the robot's vehicle controller is normal

```
yahboom@VM: ~
yahboom@VM: ~ 153x23
eader_id: 0x001(6), subscriber_id: 0x001(4)
[micro_ros_agent-1] [1785383691.095717] info | ProxyClient.cpp | create_topic | topic created | client_key: 0x00000004, topic
_id: 0x007(2), participant_id: 0x000(1) | ProxyClient.cpp | create_subscriber | subscriber created | client_key: 0x00000004, subsc
riber_id: 0x002(4), participant_id: 0x000(1) | ProxyClient.cpp | create_datareader | datareader created | client_key: 0x00000004, datar
eader_id: 0x002(6), subscriber_id: 0x002(4)
[topic_test-7] [INFO] [1785383694.681285170] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383694.681691086] [topic_test_node]: [/imu] window: 97 msgs (9.70 Hz), total: 97 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383694.681854670] [topic_test_node]: [/scan] window: 49 msgs (4.90 Hz), total: 49 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383694.681997063] [topic_test_node]: [/camera/image_raw] window: 9 msgs (0.90 Hz), total: 9 msgs, last received: 0.1 s ago
[topic_test-7] [INFO] [1785383694.682135535] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383704.681323438] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383704.681566669] [topic_test_node]: [/imu] window: 275 msgs (27.50 Hz), total: 372 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383704.681720476] [topic_test_node]: [/scan] window: 128 msgs (12.80 Hz), total: 177 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383704.681865169] [topic_test_node]: [/camera/image_raw] window: 17 msgs (1.70 Hz), total: 26 msgs, last received: 0.1 s ago
[topic_test-7] [INFO] [1785383704.682028663] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383714.681223718] [topic_test_node]: ===== Topic Data Statistics =====
[topic_test-7] [INFO] [1785383714.681471276] [topic_test_node]: [/imu] window: 273 msgs (27.30 Hz), total: 645 msgs, last received: 0.0 s ago
[topic_test-7] [INFO] [1785383714.681632226] [topic_test_node]: [/scan] window: 124 msgs (12.40 Hz), total: 301 msgs, last received: 0.1 s ago
[topic_test-7] [INFO] [1785383714.682161563] [topic_test_node]: [/camera/image_raw] window: 20 msgs (2.00 Hz), total: 46 msgs, last received: 0.3 s ago
[topic_test-7] [INFO] [1785383714.682376702] [topic_test_node]: ===== Topic Data Statistics =====
^A
```

